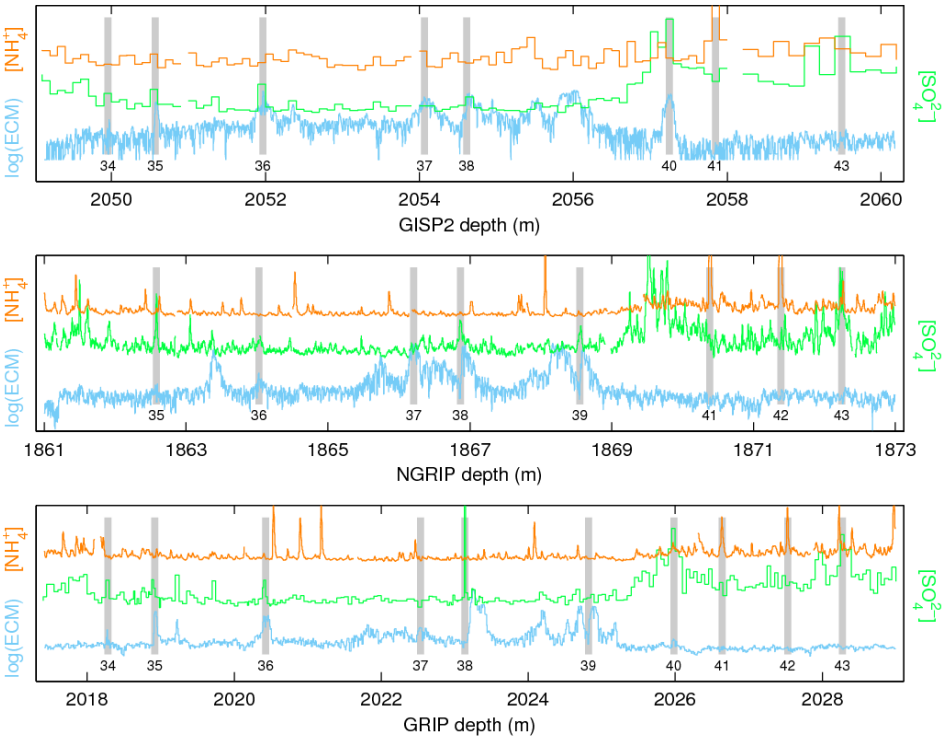
Matchmaker

A MatLab tool for manual matching of (palaeoclimate) time series



User's guide

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Introduction

Matchmaker is a collection of MatLab scripts that form an application aiming at facilitating efficient and quality-checked matching of time series from different records. It is also convenient for browsing and inspecting large data sets with many parallel data series from one record.

It has been designed for use with ice core profiles, and some design issues clearly reflect this, but can essentially be used with any type of data, although the tool is probably not suited for working with data sets from many independent cores/sites.

Copyright issues

Matchmaker can be used free of charge for any private and academic purposes. Commercial use or redistribution is not permitted without prior permission.

You are free to edit the source code to fix malfunctions, add features, or change details in layout or user interface, but if you think that your changes represent an improvement to all (or even most) users, I suggest that you forward your suggestion to me for inclusion in future versions instead of changing your local version.

I kindly ask all users to register by e-mail to me in order to be able to provide updates and to get an overview of the user group.

Matchmaker is not foolproof (and will never be) and you may still be able to provoke errors that will cause loss of data. Therefore, please act responsibly and back up your match point files frequently. No liability is accepted for loss of data or other problems.

Technical issues and preparation

MatLab version

Matchmaker has been designed to work with MatLab 7.0 (R14) or newer. A MatLab 6.5 (13) version is available, but is considerably slower when working with large data files. No support is given to the old version.

Platform

Matchmaker has been developed on Windows XP/2000 Pro/7 platforms, but is known to work under Linux as well.

Files and folders

The program can run from any folder, but subfolders named *data* and *matchfiles* must be present and contain the appropriate file types. The program itself consist of *matchmaker.m* and *matchmaker_evaluate.m* located in the root folder. The files *matchmaker_sett.mat* must be present as well. Additionally, the standard distribution contains the example file *files_main.m*, an empty match point file *empty.mrk* in the *matchfiles* folder, and a template file for building Matchmaker data files *make_data_file.m* in the *data* folder.

Data format

An example of a MatLab script for building Matchmaker data files is provided. The format is designed to handle that most ice core data are available on common depth/time scales, and that it therefore often is unnecessary to save a depth/time scale for each data series.

In short, the data file must be an internal MatLab data format (.mat) file containing the following variables:

data	cell array with N data series.
depth	cell array with M depth (or time) scales.
depth_no	vector of size N. Element number i tells which of the cells in "depth" that contains the depth scale belonging to the ith of the data series.
species	cell array with the N names of the data series.
colours	Nx3 array with RGB colour definition for the plotting.

Match point file format

Match points are saved in .mat files in the matchfile-folder. The files must contain the variable "mp" that has 2 columns: depth (or time) and match point type (see below).

The file name must exist prior to opening Matchmaker. To create a new file, copy and rename the file "empty.mat".

It is important to note, that although one match point file is needed for each data file, the match files of a certain matching project belong together. Be careful to keep track of which files are used: if 3 records are matched up, nothing prevents you from opening Matchmaker with only the first two records and remove or add match points. However, if you modify the number of match points, the third record will no longer be matched to the two others, because Matchmaker assumes that all three match point files contain parallel ordered lists of match points that are valid in all three cores.

To prevent mistakes, it is recommended to use a naming convention where the core name and a suitable name of the matching project form the match file names. For example, in a project where three cores are matched, the following names would make sense: 'CORE1_core123match.mat', 'CORE2_core123match.mat', and 'CORE3_core123match.mat'.

Letting Matchmaker know which files to use

You need to supply Matchmaker with a list of which cores to work with (names of cores + data and match point file and path names). Use the file "files_main.m" as an example and be aware of the numbering used.

Starting Matchmaker

The user can freely select how many cores to show and how many parallel data series to show for each core.

Matchmaker is started by issuing the e.g. following command in the MatLab prompt:

```
matchmaker('files_main', [1 2 3], [2 3 4]);
```

In this example, the cores with number 1, 2, and 3 in "files_main.m" is used, and the core data panels open with 2, 3, and 4 data axes, respectively.

Starting up takes a significant amount of time when the data files are larger than 50MB or when many match points have been set. Don't touch MatLab while starting Matchmaker, as

there is a real risk of making call-backs from the user interface before the call-backs have been properly defined and the settings saved.

Errors

Errors occur, although care has been taken to eliminate critical errors. MatLab warnings also occur on a regular basis.

In case of critical errors or warnings that sound worrying, cut-and-paste the error message from the MatLab prompt and send it to me by e-mail with a description of the action that caused the error to happen.

Most errors are produced within a call-back, and often the application will be able to continue without problems. However, it is probably smart to save the match points and restart Matchmaker.

Closing down after critical errors

Following a full-fledged crash, it can be impossible to get MatLab to close the windows of Matchmaker or the evaluate window. The brute-force close-down procedure is this:

"delete(gcf)"

This command closes the most recently accessed figure without any warnings or undo options. For example, be very careful not to kill the Matchmaker window (without saving) by accident if you are trying to kill the evaluate window. It is *not* a pretty solution, but better than Ctrl+Alt+Del.

Using Matchmaker

When the Matchmaker window has successfully opened, you start by choosing which of the data series to view and which depth/time interval to view them in (see section about scaling the axes). You are now ready to look for common features in the different cores and match the records using mouse clicks, and evaluate the results by letting Matchmaker stretch the data according to the match points or by using evaluation tools provided in a separate window.

Five types of match points can be set in the standard version of Matchmaker, of which the first two represent "real" match points, and the last three are used in the process of matching or when matching more than two records. Other match point types can be added upon request.

Type 1 match points (thick grey bars with numbers) are set at depths where common features are observed in the different records. The type 1 match points represent features that should be used to stretch the profiles. When pressing the *accordianize* button, the data series are stretched horizontally so that the first and last type 1 match points on the screen line up vertically.

Type 2 match points (thin grey bars) are match points representing finer, less certain match points, and are usually used to mark features or patterns that are likely to be synchronous. They play no role for the stretching of the data series.

Type 3 match points (thick red bars with numbers) are intended as temporary versions of type 1 match points. Often the matching is initially performed on long data sequences inspected in poor detail. Type 3 match points are set to mark likely match points, and a second inspection using high resolution data is then undertaken to replace the temporary match point with a type 1 match point (or to delete the match point). Type 3 match points are equivalent to type 1 match points when accordianizing or evaluating.

Type 4 match points (thick blue bars with numbers) are only used when matching 3 or more records. If a feature is common to some, but not all, records, a type 4 match point is used instead of a type 1 match point as a "dummy" match point in the record(s) that do(es) not contain the feature in order to maintain correct numbering of the following match points. Type 4 match points are numbered and are equivalent to type 1 match points when accordianizing, but are ignored when evaluating.

Type 5 match points (thin blue bars) are dummies for missing type 2 match points just as type 4 match points are for type 1 match points.

Buttons at the bottom of the screen (from left to right)

The ***Save button*** saves the match points (surprise, surprise). Before saving, the previous version of each of the match point files is copied to the same name with a ".backup" extension. If Matchmaker crashes while saving, you should be able to restore the previous version by simply renaming the file.

The ***Accordianize button*** stretches the data series horizontally to align type 1/3/4 match points vertically. The number specified in the small text field next to the button is the *leading data panel*. The accordianization uses the type 1/3/4 match points in this panel and scales the remaining data panels according to the following rules:

- If there are two or more type 1/3/4 match points visible in the leading data panel, the remaining axes are scaled so that the first and last of the type 1/3/4 match points are aligned.
- If there is only one type 1/3/4 match point visible in the leading data panel, the other axes are scaled so that the corresponding match point is between the cores, and so that the interval shown has the same length as in the leading data panel.
- If no type 1/3/4 match points are visible, nothing happens.

The number of the ***leading data panel*** or *leading record* is entered in the small text field between the *Accordianize* and *Evaluate* buttons. Counting starts from the lowest data panel.

The ***Evaluate button*** opens another figure window (if at least two match points are visible in the leading data panel, and these match points have been set in the other records as well). The window can be left open or closed after use. If left open, a second press on the *Evaluate* button in the Matchmaker main window updates the match point data and brings the *evaluate window* to the front without changing the axes scaling.

On the left graph, the depth differences between match points are plotted versus the depths of the match points in the leading record. The numbers are normalized to zero mean for the section shown. This plot can be replaced by a depth-versus-depth plot by toggling the *Depth diff* button. In both plots, the large filled dots are type 1 match points, and the large open dots are type 3 or 4 match points. Type 2/5 are filled/open small dots that are not connected by lines.

The right plot shows the relative difference between neighbouring match points (i.e., the slope of the matching curve). Inaccurate or erroneous match points can often be spotted in this panel as an interval with anomalous slope.

The x-axis scaling (i.e. the match point interval) can be changed using the text fields below the graphs. The small middle field is the step size used when browsing through the match points using the keyboard arrow keys.

Zooming can be done using the built-in MatLab zoom-function in the menu bar. If you want to return to automated scaling, right-click on the axes and choose "reset to original view". Alternatively, close and re-open the evaluate window from Matchmaker.

When clicking on a match points on the left graph, a dashed line is plotted in the corresponding position on the right graph, and the match point number (or depth for type 2/5 match points) is shown in the *Match point ID* text field.

The ***Mark button*** controls whether match points can be set or erased by clicking the axes. Type 1/2 match points are placed by clicking the left/right mouse button. Type 3 match points are set by pressing *Shift* on the keyboard while left-clicking. Type 4/5 match points are set in the same way as type 1/2 match points with the ***Dummies?*** pressed. Match points are removed if clicked.

The **Others?** button is used when comparing two sets of match points (see the dedicated section below).

The ***2nd order button*** is used to toggle display of 2nd order match points (type 2 and 5) on and off. 2nd order match point can only be set or removed when the button is toggled on.

The ***Check mps button*** counts the number of match points and presents deviations between records in a table at the Matlab prompt. The smallest distance between neighbouring match points is also shown, as very proximal 2nd order points can be very hard to distinguish on the screen.

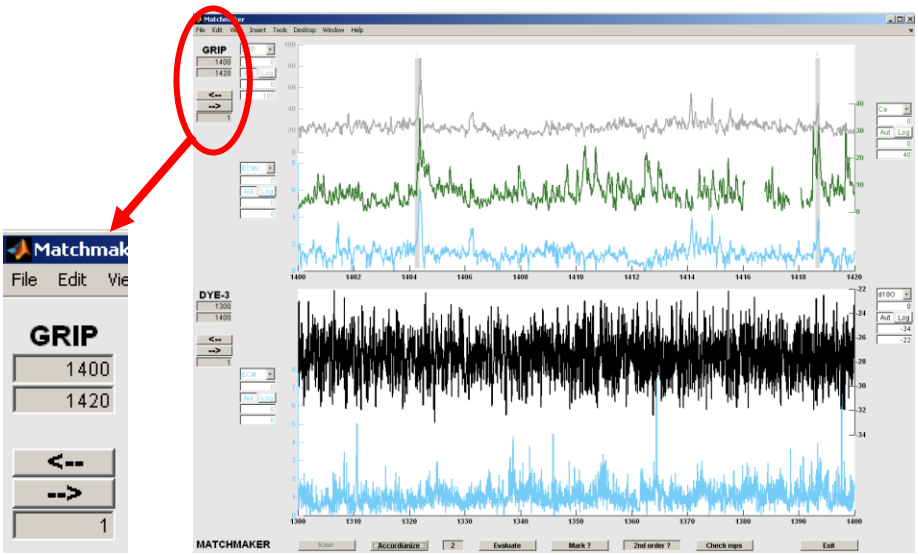
The ***Exit button***, Windows' x in the upper right corner, and *Alt-F4* closes Matchmaker. If you are trying to exit without saving changes to the match points, you are asked twice to confirm.

Keyboard commands

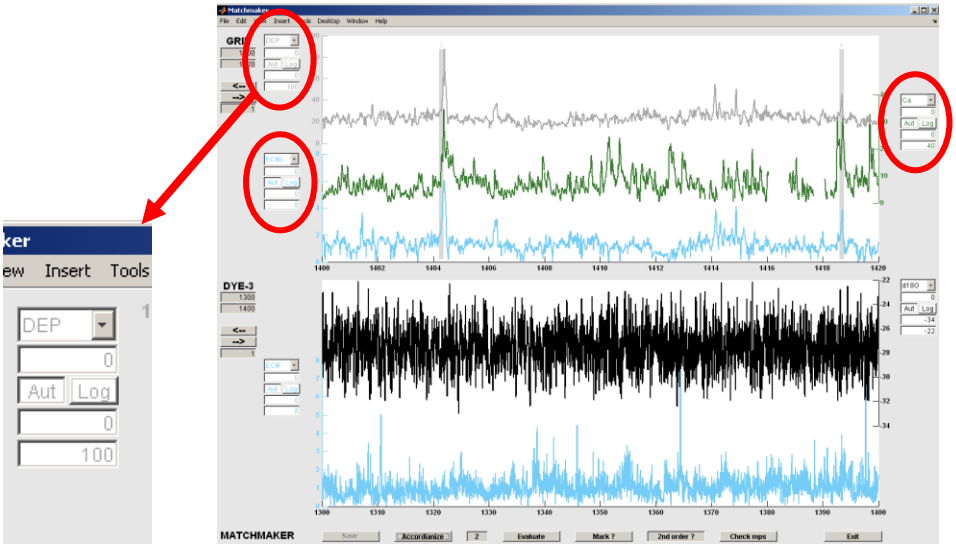
Press space bar (or any undefined key) to see a list of keyboard shortcuts (A, C, D, M, O, S, X, 2, ←,→). The arrow keys and the "C" have a special function, but apart from this, the keyboard shortcuts are similar to their respective buttons (A = Accordianize, D = Dummies, o = Others, M = Mark, S = Save, X = Exit, 2 = 2nd order). The "c/C" shortcut changes the cursor type between crosshair and full crosshair.

Pressing the arrow keys will move the x-axis of the leading data panel right or left by the distance given below the arrow buttons in the leading data panel. Subsequently, the other axes will be accordianized. This feature is particularly useful when browsing through and inspecting a set of match points.

Scaling the axes



Buttons for scaling of the depth axes are located to the left of each data panel. The field below the arrow buttons contains the increment used when using the arrows (keyboard arrows or buttons) to move the axes.



Each data sub-panel has a cluster of buttons.

The drop down list enables the user to change which data series is shown. The field below the drop down menu is a depth offset value used to shift the corresponding data series relative to the other data series (and relative to the depth scale). The unit is the same as that used by the depth axis.

The lower two fields are for manual y-axis scaling. The *Aut* button re-enables automatic scaling after manual scaling has been applied, and the *Log* button toggles logarithmic y-axis on/off.

Comparing two sets of match points

If you want to compare or review two parallel sets of match points from the same core, you can either open Matchmaker twice with the different files and settings, or instead display and edit the two sets of match points on one screen.

To do this, the two sets of match points much be merged. Load both files into MatLab and merge the files, adding the value 10 to the type of the set of match points that you regard as the secondary set of match points.

Here is an example:

```
load('myownmps.mat')           % Load match points into the variable “mp”
myown = mp;
load('theothermps.mat')        % Load the secondary set of match points into “mp”
others = mp;
others(:,2) = others(:,2)+10;  % Add 10 to second column (i.e. the type)
mp = [myown; others];          % Merge the two sets of match points
[~, idx] = sort(mp(:,1));      % Sort them according to depth
mp = mp(idx, :);
save('newname.mat', 'mp');
```

Then open Matchmaker with the new files and press the ***Others?*** button. The primary match points will be reduced in height and the secondary set will appear in the lower half of each core window. Match points are modified as usual.